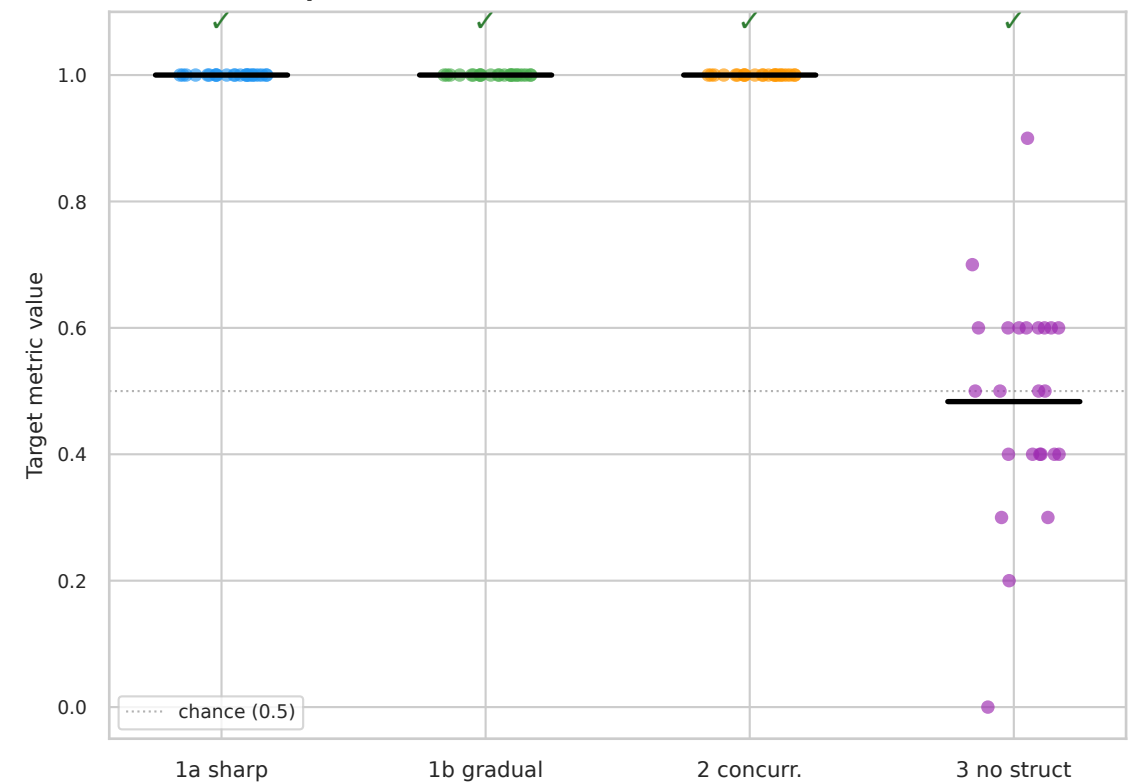
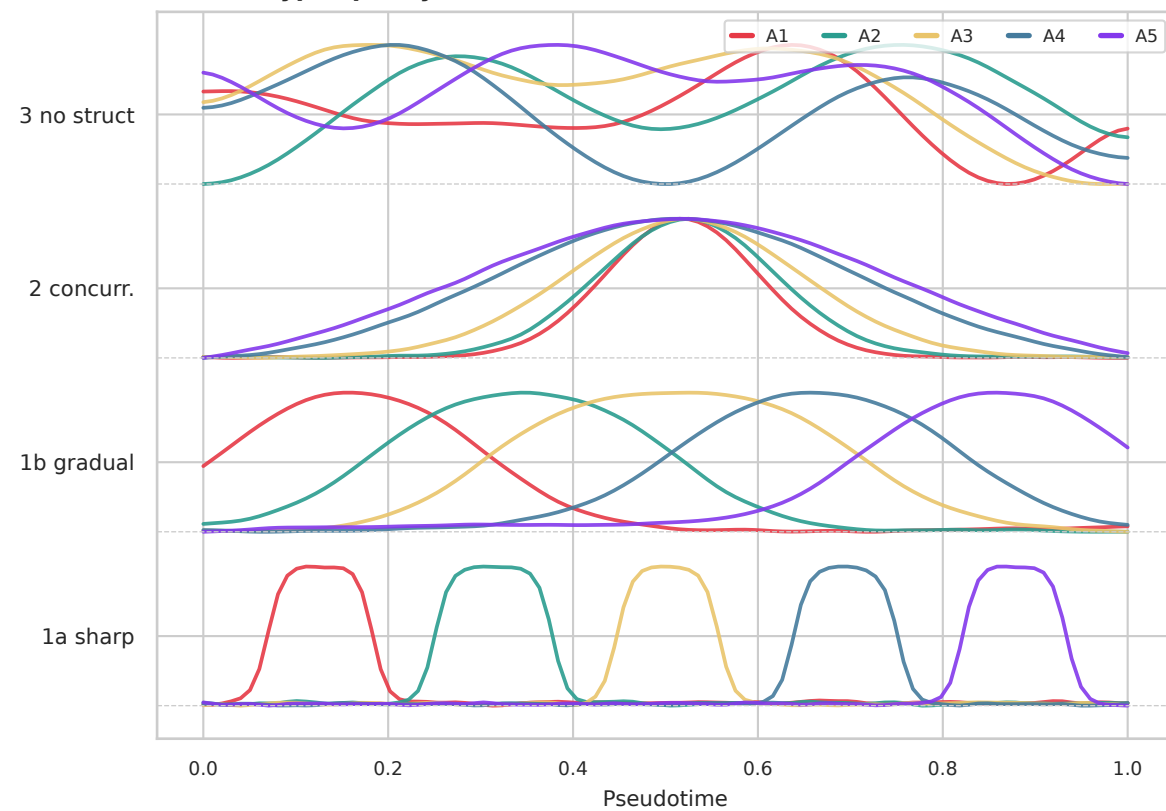


Figure 2 | Validation of the shape-agnostic peak-timing coordination metric

a | Synthetic activation profiles
(5 archetypes per system, one seed)

Synthetic positive-control benchmark — metric validation only
(24 seeds, 200 null replicates; decomposition recovery in Figs 3-4)

b | Peak-timing target metric
(across 24 independent seeds)



c | Real vs null: target metric per system
(♦ = real mean across seeds; bars = null mean $\pm$ SD)

d | Empirical p-values: real vs null
(target metric per system and null model)

$\alpha = 0.01$ (min possible $p = 0.00498$)

